

## Determinants

$$\det A = |A| = (-1)^{\text{row swaps}} \times (\text{product of pivots of } U)$$

↳ eliminate, count the swaps.

### Properties:

1.  $\det I = 1$ ,  $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$

2. A row swap flips the sign,  $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$

3a. scaling one row by  $t$ , scales det. by  $t$ .

$$\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

3b.  $\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$

linear combinations

4. Two equal rows or zero row,  $\det A = 0$ :  $\begin{vmatrix} a & b \\ a & b \end{vmatrix} = 0$   $\begin{vmatrix} a & b \\ 0 & 0 \end{vmatrix} = 0$

### Singularity:

$\det A = 0$  is equivalent to:

$A$  is singular,  $\text{rank} < n$ , dependent cols,

no  $A^{-1}$ ,  $N(A)$  contains non zero vector...

Gauss-Jordan to find  $A^{-1}$ :

$$[A \mid I] \rightarrow [I \mid A^{-1}]$$

When  $\det A = 0$ , it means one of the pivots must be 0.

### From properties 1-4:

• row subtraction changes nothing:

$$\begin{vmatrix} a & b \\ c-a & d-a \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \underbrace{\begin{vmatrix} a & b \\ -a & -b \end{vmatrix}}_{(3b.)} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} - \underbrace{\begin{vmatrix} a & b \\ a & b \end{vmatrix}}_{(3a.+4.) = 0}$$

• for triangular matrix  $\det A = \text{product of diagonal}$ :

$$\begin{vmatrix} p_1 & x & x & x \\ 0 & p_2 & & \\ & & \ddots & \\ & & & p_n \end{vmatrix} = \begin{vmatrix} p_1 & 0 & 0 & \\ 0 & p_2 & & \\ & & \ddots & \\ & & & p_n \end{vmatrix} = p_1 \times p_2 \times \dots \times p_n \quad \begin{vmatrix} 1 & 0 & 0 & \\ 0 & 1 & & \\ & & \ddots & \\ & & & 1 \end{vmatrix}$$

(elimination rule) (from 1 + 3a.)

Det. vs cond.:

Det. scales with  $t$ ,  $\det(tA) = t^n \det(A)$ .

Even with singular matrix if  $t$  is small, det. will be close to 0.

To detect singularity use  $\text{cond}(A) = \frac{\sigma_{\max}}{\sigma_{\min}}$  if  $\text{cond} = 1$  means invertible.